

INSTITUTO SUPERIOR TÉCNICO

Departamento de Engenharia Informática

Forensics Cyber-Security

MEIC, METI

Digital Forensics Report

Lab Assignment II – OPERATION DEEPFISH – Stage II

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1 Acquired artifacts

Every artifact below is identified by its SHA-256 (the complete manifest of submitted files is in Appendix D) and by a reference ID (`A1xx`) used throughout this report. Evidence images were verified against the hashes recorded before analysis. Derived items (the decrypted volume, extracted files, keys, hive copies) are working copies, not evidence, and are kept out of the submission.

ID	File	SHA-256
A101	windows_laptop_memory.raw	c0b66a9ec5ff16d8c01726a8717648f729857fa059ba7485c5c1717587892be3
A102	windows_laptop_disk.dd	5eca1f65df49ccf5bddeb8c5c3a44fe1a8a9e6faab04ed601d565b915bbaa6a8
A103	linux_workstation_disk.dd	b31261098d901de0be201f4f78f5816c946490b9eb4ae2c94dc2fcf4af4e3368
A104	handwritten-note-fig1.png (Figure 1)	d259e34bfc965a1aea720ffa7b9443ebfde54b6fdb78d26811fd5c44149b6896
A124	stash.zip (Linux, `backups/`)	d0544dcbcbccbce302ad5f811de9f59ee786d94e7b311b2d3ae6f3b153a0c308
A132	server.js (Linux, `backups/peixinho/`)	2a014a1168968989cf36488ac8a7f218f8b3591dabeea50199c6f26e03abf863
A133	peixinho .git/config (Linux)	dd138156547b37e8615e3d4b8acd593958c63aa3f3b2367f247192f766d4cde5
A149	acme-challenge token (Linux repo)	69296b1d98e75e397d885f5c9caef8650676094188e05484b57011ba2ea11643
A137	Security.evtx (C:)	29e6a0df28080267b2e470e946a202dd1ff27598b6deedbd593b7e7ad49ba36f
A139	PowerShell Operational.evtx (C:)	2467ae88d623459f9244ad633b3943feb8adcf73b67fac63eac9408c307739a7
A140	System.evtx (C:)	1da2c58c5738b96c86f40c17fee02e11f26b70e2423e71e37d682c0c36093126
A138	WinRM Operational.evtx (C:)	3e368494f50bf4cc35d1bb31bbe7eb91217c9d2eb94fb7311c36381ec6706872
A141	WINRSHOST.EXE prefetch (C:)	2a5e57c55a5a5692a0a90ae42ee0a2ff0f65e1495ed4d1ec5459908cf29ff25e
A142	POWERSHELL.EXE prefetch (C:)	fbf8f0b5ff58fea3a8d0d11d37de4ae4de6532cca266b5952ef3bec29b0ef340
A143	setupapi.dev.log (C:)	c026d59a4c79d211bdf6fa6b0a40d203780595a746677fc6cb0ff9e1fb2238a6
A152	mom-photo.jpg (D:, note part 3)	fe8f99ce75f92a4aee463d5b535d2737e0e042d8bfb59bf44d123c9704f6be55
A153	hate-this-place.jpg (D:, note part 5)	c323d401bd25fd8a6ae6e3b8f7dba724ce95d51c886b15653a2f5a0863c800c8
A155	ConsoleHost_history.txt (D:, PowerShell)	53dcd531d211c2a219df6ff859357c9ce018296e53106a0b097c9d3356314372
A156	edge_History / operagx_History (D:)	34977f57c1f75b5bff31d71ccb5db189265635bb4bbf28d0c0ae97633a80d165 / 3890e8ed751cb39632ee7a73461db812745b98e768a40b3ba7b9717aee15cf8e
A157	Discord cache (D:, 9 blobs `f_000001-9`)	nine SHA-256 in the manifest (Appendix D)

The eight source artifacts of Table 1 and the hidden files of Table 2 of the assignment were checked against the seized systems as described in Section 3.1. The complete SHA-256 of every file submitted in the ZIP is listed in Appendix D, and the exact commands that recovered the main artifacts are in Appendix E; the three seized images (A101–A103) and `stash.zip` (A124) are too large for the archive and are referenced by hash only.

2 Report of all findings (Investigation and Evidence Analysis)

Summary.

1. The Windows laptop's BitLocker-protected D: volume was decrypted with the volume's FVEK, recovered from memory (A108, A107). The key was validated on the NTFS boot sector and the $MFT record.

2. The password of the handwritten note is `johnny-pc_elliot_shibuya_fsociety_pimenta`. It opens `stash.zip` from the Linux workstation (A124, A125). The five parts are confirmed together by the archive's CRC check. The Windows password `fsociety` is confirmed independently by the SAM hash (A118).

3. Note parts 3 and 5 name a region shown in a *Jujutsu Kaisen* picture and a restaurant in a family photograph. Neither word is written in, or steganographically hidden in, the two images recovered from the D: volume (`mom-photo.jpg`, `hate-this-place.jpg`; in-band steganalysis negative, A154). They were identified by reverse image search: `mom-photo.jpg` is a scene from the "Shibuya Incident" arc, set at Shibuya Station (part 3 = `shibuya`), and `hate-this-place.jpg` is the restaurant *Maria Pimenta* at the Fábrica da Pólvora de Barcarena (part 5 = `pimenta`). Both readings are corroborated cryptographically, as the two words are among the five parts whose SHA-256 opens `stash.zip` (A152, A153, A125).

4. The Linux workstation holds a copy of the phishing server source (`server.js`, same hash as Stage I A006), a git clone of the phishing host (`fenix.dpss.inesc-id.pt`), the phishing campaign's acme-challenge token (A149), and an encrypted archive of seven of the eight Stage I source artifacts (A126).

5. Shell history on the Linux workstation records the steps used to hide the Stage I files inside media (A129), the encryption of the archive, secure deletion (`srm`) of the working directories, and a `git rm` and commit on the production server `fenix-prod` (A131).

6. On the Windows laptop, a PowerShell script block copies files into place and resets their timestamps from a supplied table, that is, timestomping (A139). It ran over WinRM on 22 August 2026 (10:29 WEST), the last day of the reported phishing window.

7. The PowerShell operational log was cleared on 28 July 2026 by the account `johnny` (A140), in the same second that `background.jpg` was deleted (E2).

Timestamps are in WEST (UTC+1). Linux image timestamps have the assignment's 27-day correction applied, and each such row says so. Windows timestamps need no correction. The full timeline is in `report-and-timeline/timeline-events.csv` (rows E1–E20, T1–T3, M1–M17).

Methods and tools. Sleuth Kit (`mmls`, `fsstat`, `fls`, `icat`, `istat`, `jls`, `jcat`); Volatility 3 2.28.2 with `windows.bitlocker` (third-party plugin, reviewed and SHA-256 checked before use); `cryptography` for AES-XTS decryption; `pyevtx` (libevtx) for event logs; `pyscca` for Prefetch; Impacket `secretsdump` for offline SAM and LSA parsing; `foremost` for carving; `exiftool`. Scripts are in `results/scripts/` and every step is reproducible from `kali.sh`. Symbol tables for the memory image were obtained over the network (`-net`); only the public kernel PDB identifier was sent.

3 Analysis of relevant findings

3.1 Phishing website and recovered artifacts

Question 1. *Do the seized systems contain traces of the phishing website and/or the hidden files recovered during Lab Assignment I?*

Observations.

- *Linux workstation.* `home/johnnymusk/backups/peixinho/server.js` has SHA-256 `2a014a11...` (A132), the same hash as the phishing server recovered in Lab 1 (A006). Its `.git/config` names `https://fenix.dpss.inesc-id.pt/.git/` as the remote (A133), which is the host of the phishing page in Lab 1. Its reflog records a single clone of that host on 23 August 2026 (A134, E14).
- *The acme-challenge token.* The repository contains `.well-known/acme-challenge/9xQ3...` with the same token as Lab 1 A005. The bytes differ only in the line ending (LF here, CRLF in A005) (A149).
- *Stage I source artifacts.* `stash.zip` (A124) holds nine files; seven of them match the Table 1 hashes of the assignment (A126): `myzip.zip`, the anime `.mkv`, the GTA iceberg image, the GTA trailer, the `dj_cara` WAV, `blank` and `polish_cow.webp`. The telemetry workbook is in the archive with the same values as the published file but a different hash (A127, A128). The source sample named `ESP32_ADXL345_3Axis_Telemetry.xlsx` is therefore not byte-identical to the published version.
- *Windows laptop.* No Stage I file is present as a live, readable file on C: or on the user area of D: (A113, A146). The `Users/johnny/Pictures` folder on D: holds two images (A111, A112). Neither is a Stage I file.
- *PowerShell history on D:.* `ConsoleHost_history.txt` (A155) records `sshmuskyboi@146.193.41.109"cat~/fenix/peixinho/credentials.txt"`: the laptop user connecting to the production host named in the SSH config and `deploy.sh` and reading the harvested `credentials.txt`.
- *Recovered Discord chat on D:.* The client cache (A157) holds a conversation in which the laptop user describes running the phishing page, capturing a TA's credentials, and both receiving and hiding the set of files later found embedded in the media artefacts — a first-person account of the traces above (developed in Section 3.4).

Interpretation. The phishing kit (server source, acme token and git remote) was stored on the suspect's personal Linux workstation, and that workstation was used to package the Stage I files. The PowerShell history additionally ties the Windows laptop to the *operation* of the live site: the user read the campaign's harvested `credentials.txt` on the production host (A155). The Lab 1 hidden files (Table 2) were not found as standalone files. The shell history indicates they were embedded into media files with encoding scripts (A129). Five of the eight hidden files are verified by hash: f1.png, f2.json, amelia.pdf and f8.webp against the Lab 1 recovery (A150), and f4.pdf carved from the GTA trailer (A151). The remaining three (f5.py, f6.json and f7.diff) are embedded with bit-level encoders whose scripts were removed (A129), so they are not verified.

3.2 Provenance and temporal reconstruction

Question 2. *Can you reconstruct the provenance and handling of these files?*

The table lists the events that bear on provenance. Windows times are native; Linux times are corrected by -27 days. All times are WEST.

Time (WEST)	Event	Source
2026-04-15 21:17	Earliest commit in the `peixinho` repository	M1
2026-07-28 21:16	PowerShell operational log cleared by `johnny`	T1, A140
2026-07-28 21:16	`background.jpg` deleted on D:	E2, A122
2026-07-28 22:44	`hate-this-place.jpg`, `mom-photo.jpg` created on D:	E3, A122
2026-08-19 to 22	Reported phishing campaign window	M14, M15
2026-08-22 10:29	Timestomping script executed over WinRM	T2, A139
2026-08-22 10:30	`winrshost.exe` and `powershell.exe` run	T3, A141, A142
2026-08-23 23:51	`peixinho` repository cloned from `fenix.dpss.inesc-id.pt`	E14, A134
2026-08-25 13:01	Published `Telemetry.xlsx` copy saved	E9
2026-08-25 15:06	Telemetry copy inside `stash.zip` saved (embedded date, corrected)	E10
2026-08-25 15:56	`stash.zip` created (`zip -e`)	E18, A129
2026-08-25 22:02	WinRM session from `10.0.2.2` port 5985 (memory, state CLOSED)	M17, A101
2026-08-26 10:29	Shell history last written	E20, A129

Observations. The archive was created on 25 August, using files from the hidden-file staging folder, and encrypted with `zip -e` (A129). The archive password was derived from five values that all appear in evidence (Section 2). The repository was cloned on 23 August, the day after the reported window (19–22 August) had ended, and before the archive was built. The timestomping script ran on 22 August, within that window.

Interpretation. The sequence is consistent with: clone of the phishing host (23 Aug), staging and encoding of the Stage I files and packaging of the archive (25 Aug). The timestomping on 22 Aug is more likely part of the phishing-period activity than of the staging, but the logs do not show which files it injected.

3.3 Anti-forensic activity

Question 3. *Did you find evidence of anti-forensic activity?*

Yes. The following techniques were observed. Each is recorded with its evidence.

1. **Timestomping.** A PowerShell script block (`C:\Windows\Temp\Modules\Forensics.psm1`) copies files into a destination and sets their creation, write and access times from a supplied table (A139, T2). The module file itself is not in the live listing or among deleted entries (A139).

2. **Event-log clearing.** The PowerShell operational log was cleared on 28 July by `johnny` (A140, event 104, T1). The Security log has no clearing event. Other logs may have been cleared without a record.

3. **Secure deletion.** The Linux history shows `srm -ll -r` on the working directories, after installing `secure-delete` (A129). Both directories are absent from the live filesystem (A103). The ext4 journal no longer holds their names (A147).

4. **Concealment in media.** The Stage I files were embedded into media files with purpose-built encoding scripts (LSB, DCT, polyglot and WAV encoders; A129). The browsing history for the same days covers those techniques (A135).

5. **Encrypted archive.** The archive was created with `zip -e`, and its source files were moved into it with `zip -m` (A129, E18, E19).

6. **Remote history rewrite.** On the production host `fenix-prod`, the suspect ran `git rm -r` on the site content and committed it as "cleanup" (A129). The local bundle of the repository contains that commit (M13).

7. **Stated intent to destroy evidence.** In the recovered chat (A157) the user plans to throw the USB holding the supplied files into the Tejo and to keep the material offline and unopened. This is a

statement of intent, not an observed on-disk action, but it is consistent with the secure-deletion already carried out.

Limitations. Anti-forensic actions that left no record cannot be identified. The Prefetch files do not record the 25 August run that memory shows, so Prefetch and memory disagree on that point and this is not explained.

3.4 New discoveries and investigative interpretation

Question 4. *Do these findings strengthen or weaken the attribution of the phishing attack to João Musk, or point to other relevant actors?*

Recovered communications (A157). A Discord conversation (144 messages) was recovered from the client's local cache on D: (`Cache_Data/f_00000N`, gzip blobs decoded to JSON; one blob is a voice note). In it the laptop's user, `Johnny`, states that he ran a phishing page based on the IST password-reset e-mail, captured a Natural-Language teaching assistant's credentials, and used them to pull that TA's exams from the TA's drive — which matches the credential read (A155) and the `scp` of the exam (A155) found independently on disk. Separately, a long-standing online contact, `DuckyDonald`, who says he interns at `AMA` on AI work, asks `Johnny` to keep a set of files safe, leaves them on a USB drive in a public locker ("Locky" locker, Alameda Metro, number 3, code 676767), and instructs him to hide each file with the supplied encoding scripts, store them in the password-protected archive and secure-delete the originals. `Johnny` does so, then realises the folder auto-commits and pushes to git, so the hidden files were published and reached his site through the merge — which matches the exposed `.git` and the `peixinho` history. `DuckyDonald` then reveals that he is `Tiago Sintaxe`, the same TA whose exams were stolen, and tells `Johnny` to take the USB to the police if he disappears; `Johnny` instead says he will throw it into the Tejo.

Strengthening indicators.

- The Windows laptop belongs to the user `johnny` and the computer is `JOHNNY-PC` (A101, A137). The Linux workstation belongs to `johnnymusk` (A103). The handwritten note's first password part is the laptop name, `johnny-pc`.
- The Linux user holds the SSH identity `muskyboi` for the production host `fenix-prod` (A131). The assignment links the `MuskyBoi` accounts to the suspect.
- The user's to-do note says "back up peixinho" and points to the BitLocker-locked D: volume for an exam (A130). That volume is the one decrypted in this analysis, and it holds the `Exams_LN` folder (A122).
- The phishing kit (A132, A133, A149) and the archive of Stage I sources (A126) are both on the same workstation.
- The laptop's PowerShell history shows its user reading the campaign's harvested `credentials.txt` on the production host over SSH as `muskyboi` (A155), tying the same person to the *operation* of the live phishing site, not only to its source code.
- The laptop's Opera GX history records the planning of the campaign: searches for *how does phishing works* and *social engineering* and a ChatGPT query *how to create a phishing website* (10 Aug), then *how to send email as someone else* and a visit to the mail-spoofing service `emkei.cz` on 19 Aug, the first day of the campaign window (A156). This indicates the laptop's user actively researched and prepared the phishing site and its delivery e-mail.

Weakening or competing indicators.

- The reflog records the clone as authored by `António Alves` (A134). This may be the upstream author of the repository, but it is a second person, and the role is not established.

- The production host address in the SSH config (`146.193.41.109`) differs from the phishing host address in Lab 1 (`146.193.41.5`). The two are linked by hostname and account name only.
- Two different accounts were used for remote sessions on the laptop: `johnny` on 22 August and `Administrator` on 25 August (A137). The logs do not show who used each one.

A second access path (exam exfiltration). The same PowerShell history (A155) also records `ssh ist101337@sigma.ist.utl.pt` followed by `scp` of `Natural_Language_MAP1_2026-2027.pdf` from that host's `Linguagem_Natural/Exams/` folder into `Desktop\Exams_LN\` on D:. This matches the exam file seen in the USN journal (A122, E4) and the to-do note (A130). It shows a second account, `ist101337`, used from this laptop to copy exam material off an IST host. Whether `ist101337` is the suspect's own account or a credential obtained through the phishing campaign is not established by this evidence; both are consistent with it, and it is a lead for the Polícia Judiciária rather than a settled fact.

Assessment. The evidence supports the view that the phishing kit and the Stage I material were under the control of the suspect's personal accounts. It does not by itself establish who ran the remote sessions on the laptop or who injected files. An external actor with access to the laptop cannot be excluded on the present evidence.

Hypotheses on the wider scheme (not established). Names in the shell history (`ama_api_tokens.json`, `amelia_tools_config.json`, `rag_poisoning_patch.diff`, `train_anisotropic_override.py`) suggest that the hidden material concerns an AI assistant: its access tokens, its tool configuration, a retrieval-poisoning patch and a training script. The stash also contains a cloud service credential file (Table 2, f6). If these are the contents of the hidden files, the scheme may aim at credential theft and manipulation of an AI system, rather than only phishing. We have not verified the contents of the hidden files, so this is a hypothesis. The recovered conversation (A157) supports the view that the hidden material did not originate with this suspect: it was supplied on a USB by the contact `DuckyDonald`, who presents himself as an AMA intern and later as `Tiago Sintaxe`, and the suspect's role on these files, as described there and consistent with the disk (A129), is to hide, encrypt and (inadvertently) publish them — in contrast to the phishing site itself, which the laptop's own search history shows its user researching and preparing (A156). This is an account given in chat, corroborated by the on-disk artefacts but not independently proven as to the real identities involved.

Recommended next steps for the Polícia Judiciária.

1. Obtain the production host's logs (`fenix-prod`) for the 'cleanup' commit and for the SSH key `muskyboi`.
2. Obtain account records for the GitHub and GitLab `MuskyBoi` accounts and the upstream repository history.
3. Obtain network and authentication logs for the laptop around 22 and 25 August, including the address `10.0.2.2`.
4. Recover the module file `Forensics.psm1` and the files it injected from backups, shadow copies or the cloud, and verify their content.
5. Confirm the BitLocker recovery key through the account that owns it, and check whether the USB device `TEARSINRAIN` was ever connected to any other machine.
6. Verify the content of the hidden files (Table 2) and the contents of the encrypted archive against the hashes in Appendix D.
7. Identify the account `ist101337` and obtain `sigma.ist.utl.pt` authentication and `scp` logs around the exam exfiltration recorded in A155, to establish whose credential was used.
8. Identify the persons behind the handles `DuckyDonald` / `Tiago Sintaxe` and the group-chat members (A157); request the Discord account and server records; and seek the USB drive

described in A157 (the Alameda "Locky" locker, number 3, and the river-disposal statement) and the AMA/AMÉLIA material it held.

3.5 Limitations and open items

- Three hidden files of Table 2 (f5.py, f6.json, f7.diff) remain unverified: they are embedded with bit-level encoders, and the decoders were removed from the suspect's machine.
- Unallocated space on the Linux disk was not carved.
- Times in the Linux history and the zip members have no stated time zone; WEST was applied as recorded.
- The C: partition's third volume (768 MB, sector 41199616) was only checked at the top level.
- The Security log is intact apart from the PowerShell clearing; logs on other machines were not available.

4 Use of Artificial Intelligence

AI assistance was used in this work. An AI coding assistant (Claude, Anthropic) was used to help write and debug the analysis scripts, to assist with the steganalysis and identification of the two D: images, and to draft and revise the report text. It was not relied on as a source of facts: every finding was reproduced from the evidence, and every claim in this report is tied to a submitted evidence artifact (Appendix D) and to the command that produced it (Appendix E).

5 Appendices

Appendix A: artifact IDs. Findings are cross-referenced by ID (`A101–A157`). The SHA-256 of every submitted file is in Appendix D and the commands that produced the main artifacts in Appendix E; the team's full per-artifact working notes (observations, interpretations, limitations and inodes) are kept as an internal record and are not part of this submission.

Appendix B: timeline. `lab2/report-and-timeline/timeline-events.csv` (rows E1–E20, T1–T3, M1–M17).

Appendix C: scripts. `lab2/results/scripts/`: `kali.sh` (all steps), `bitlocker.py`, `note_password.py`, `search_volume.py`, `compare_xlsx.py`, `evtx_dump.py`, `generate_candidates.py`, `image_steg_check.py` (steganalysis of the part-3/5 photos), `extract_d_images.sh` (D: image recovery), `parse_discord_cache.py` (Discord cache decode).

Appendix D: submitted evidence (SHA-256 manifest)

Every file in the submitted `evidence/` tree, byte-exact. Paths are relative to `evidence/`; the list is in `sha256sum` format and can be verified with `sha256sum -c`. The `discord/*.json` files are the gzip-decoded copies of the corresponding `f_00000N` cache blobs (A157).

```
d259e34bfc965a1aea720ffa7b9443ebfde54b6fdb78d26811fd5c44149b6896  handwritten-note-fig1.png
dfea6a788d615caa44105fa4d0dd893f67c39a5f055f726abdb544c3e5c32f2d  linux/deploy.sh
c1e8881434f8a5d5196017cf2a185f03a5e89d03357c49c6557292121f13ffbd  linux/peixinho-git-log.txt
7c892d4340d2c7565aabbd334b8485789918e9b86d5f0ca9c2cf5ee5bf989c5a  linux/peixinho.bundle
160e0082a3130c3b7d3f068c0174832a609f677222589cc1c641668aedebb4a4  linux/ssh-config.txt
1f1069f876999cd9f37bcc783610f1f4ba1c81d783953c15a82122efcd295f06  windows/bitlocker-hash.txt
53dcd531d211c2a219df6ff859357c9ce018296e53106a0b097c9d3356314372  windows/ConsoleHost_history.txt
34977f57c1f75b5bff31d71ccb5db189265635bb4bbf28d0c0ae97633a80d165  windows/edge_History
3890e8ed751cb39632ee7a73461db812745b98e768a40b3ba7b9717aee15cf8e  windows/operagx_History
fe8f99ce75f92a4aee463d5b535d2737e0e042d8bfb59bf44d123c9704f6be55  windows/mom-photo.jpg
c323d401bd25fd8a6ae6e3b8f7dba724ce95d51c886b15653a2f5a0863c800c8  windows/hate-this-place.jpg
1a3320b8c6a507e47d43b5be05c9c852edef7010a1ba89a07cec473b383e41b6  windows/discord/discord-messages.txt
f63f1b0aa059c51f5286c93ada16bffb6af5f8fc9aaa24f1c75361b6e1bb85d0  windows/discord/f_000001
cee74887189dab0d8bd3bea37e437390bb78f639d832bf8c06426349a89c9c8f  windows/discord/f_000002
4368ecda87cef73720da384fd9167951547096ec85e7a99c7c7cb69698286f90  windows/discord/f_000003
c25c32d4c61b498624369264c216a19b2782a31868f94a7537c1cdec4fc7274b  windows/discord/f_000004
a4fda2c29a82d461a709cdc3ea4707651549a89d9782f3d1e696b81833edeccc  windows/discord/f_000005
835d330d451afb4f9d47bde6dc9f1615a24001a1a82c6a10046a306a358d2ae8  windows/discord/f_000006
bf0d6c3cd0d8eca1d5e538359ca90292b6230f53b63dd1bd17a18bdb677f38b6  windows/discord/f_000007
39f7d8562d46c0fe4ba272df373e7071684afb71654bee4080d6afa0988edc27  windows/discord/f_000008
81822b1efac7713c3d355cfd5aa4097483ac3e305aa83ad25de650077325d7a2  windows/discord/f_000009
62c1a719d456aebb2160a03f2a82b56224789360fdbdae260a6f6465cf60b359  windows/discord/f_000001.json
4a99ab9f5f194ef0a0211f7911231483db789dde1f31eb837d6ed6525c8debd0  windows/discord/f_000002.json
098873500e2ca88f946e0142717c334c2ed051bec34fe610f4e77af8b6446d35  windows/discord/f_000003.json
2072cde64c8ad0a6a47a2a8e52baa8ead2053ebab834b3ff6fc388123adc5239  windows/discord/f_000004.json
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855  windows/discord/f_000005.json
f773e14093a084bc87f5358bcc64681815db51a2b9302cb9dfbbce76f9972d8c  windows/discord/f_000006.json
809a1f1c2c6daae06540e54549998b529cd22c26d200008b36feec2a853f5fbe  windows/discord/f_000007.json
a0e5cc5ecc9df390b6b17aea91fd1ba87115f499994fa1690dda9b0d469c3a8e  windows/discord/f_000008.json
96e562fecbf620e298906a6612aedf628d1036c488b35b7172bf7a6de4dfcc8f  windows/discord/f_000009.json
```

Appendix E: key commands to reproduce

Read-only against the images; derived files are working copies. Inodes are given inline with the relevant commands below.

```
# Chain of custody: hash the three seized images first
sha256sum windows_laptop_memory.raw windows_laptop_disk.dd linux_workstation_disk.dd

# BitLocker D: -- recover the FVEK from memory, then decrypt and mount read-only
vol -p plugins -f windows_laptop_memory.raw \
    windows.bitlocker.BitlockerFVEKScan --tags FVEc Cngb None --dislocker
LOOP=$(sudo losetup -r -o 21880635392 --show -f windows_laptop_disk.dd)
sudo dislocker -v -k 0x<off>-Dislocker.fvek -V "$LOOP" /mnt/dislocker
sudo mount -t ntfs-3g -o ro /mnt/dislocker/dislocker-file /mnt/dvol

# Windows credentials (SAM+SYSTEM): johnny's password = part 4 of the note
impacket-secretsdump -sam SAM -system SYSTEM LOCAL
john --format=NT --wordlist=rockyou.txt hashes.txt     # -> fsociety

# Handwritten-note ZIP password (opens stash.zip)
#   sha256(johnny-pc_elliot_shibuya_fsociety_pimenta)
python3 note_password.py stash.zip

# Note parts 3/5: the two D: images; no steganographic payload
python3 image_steg_check.py mom-photo.jpg
python3 image_steg_check.py hate-this-place.jpg

# PowerShell + browser history (read large files via TSK, not the FUSE mount)
sudo icat /mnt/dislocker/dislocker-file 1102 > ConsoleHost_history.txt   # inode 1102
sudo icat /mnt/dislocker/dislocker-file 1525 > operagx_History          # inode 1525
sqlite3 operagx_History \
  "select datetime(last_visit_time/1000000-11644473600,'unixepoch'),url from urls;"
```

```
# Discord conversation: cache blobs are gzip -> JSON (f_000005 is Ogg/Opus audio)
for f in f_00000{1,2,3,4,6,7,8,9}; do gzip -cd "$f" > "$f.json"; done
python3 parse_discord_cache.py f_*.json > discord-messages.txt

# Verify the submitted evidence manifest
sha256sum -c SHA256SUMS
```